

AquaGuard AI

- Smart Water Waste & Leakage Risk Detection System
- AI for Sustainability Virtual Internship
- 1M1B × IBM SkillsBuild × AICTE
- SREENIVAASINI.J | College Name: PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY

The Problem

- Unnoticed leaks and abnormal water consumption can cause avoidable water wastage.
- Manual monitoring may miss unusual consumption patterns.
- Maintenance teams need early warning and location-level prioritization.

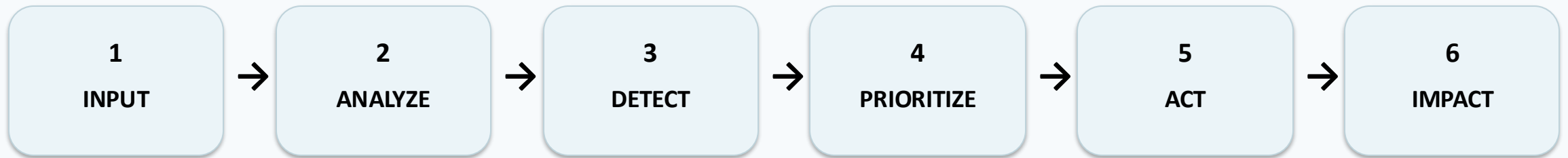
Problem Statement

- How might we use AI to detect abnormal water-consumption patterns so that water management becomes more sustainable?
- Primary SDG: SDG 6 – Clean Water and Sanitation

Solution Overview

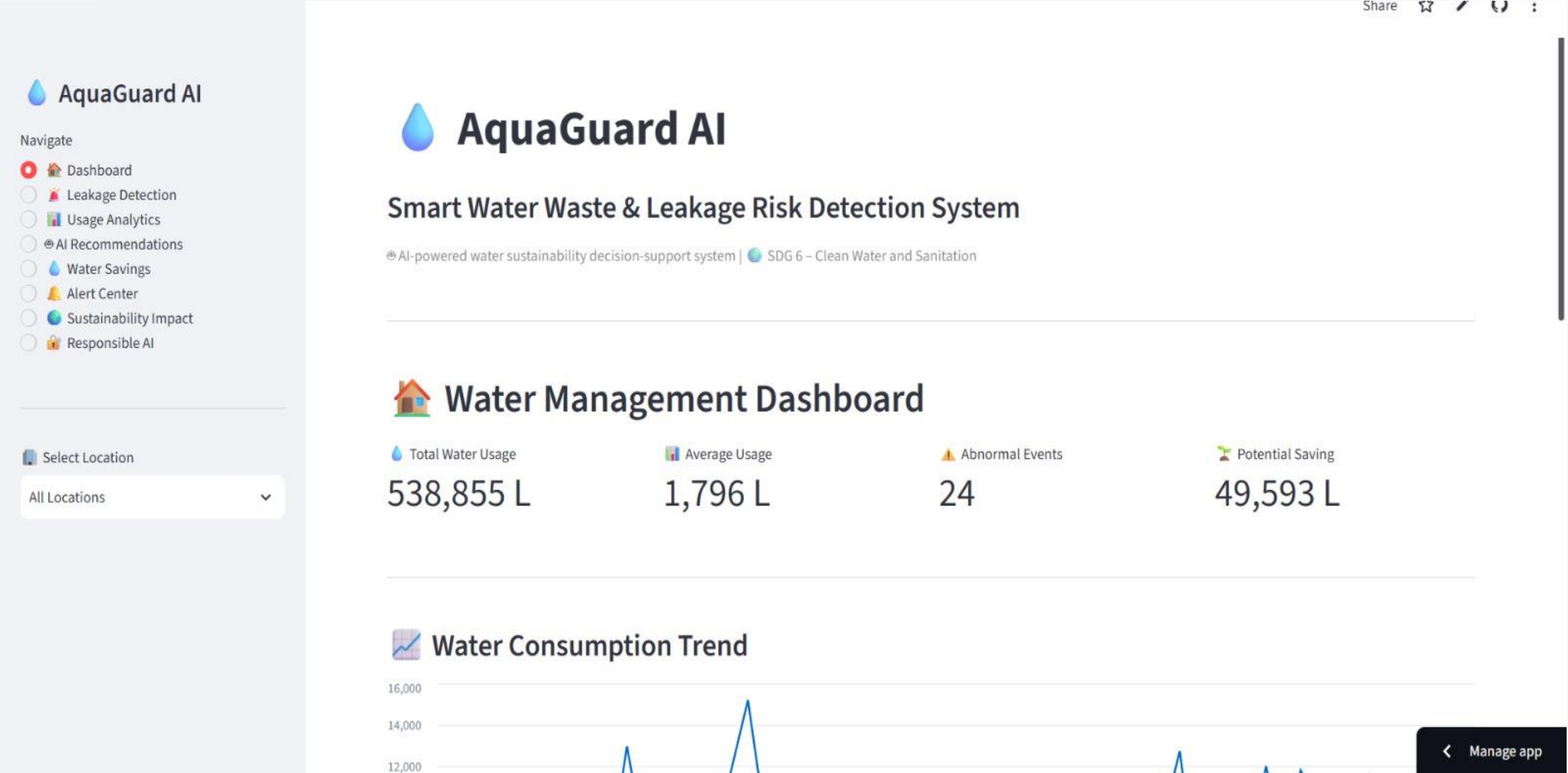
- Analyze water usage → detect abnormal patterns → calculate risk → recommend action → estimate potential savings.
- Designed as an AI-powered decision-support and early-warning prototype.

AI Workflow 🤖



Water usage + occupancy → Isolation Forest → risk indicator → alert → recommendation → potential saving

Dashboard Preview



Leakage Detection

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Navigate

-  Dashboard
-  Leakage Detection
-  Usage Analytics
-  AI Recommendations
-  Water Savings
-  Alert Center
-  Sustainability Impact
-  Responsible AI

Select Location

All Locations

 Leakage Risk Detection

Share









 High Risk

25

 Medium Risk

45

 Low Risk

230

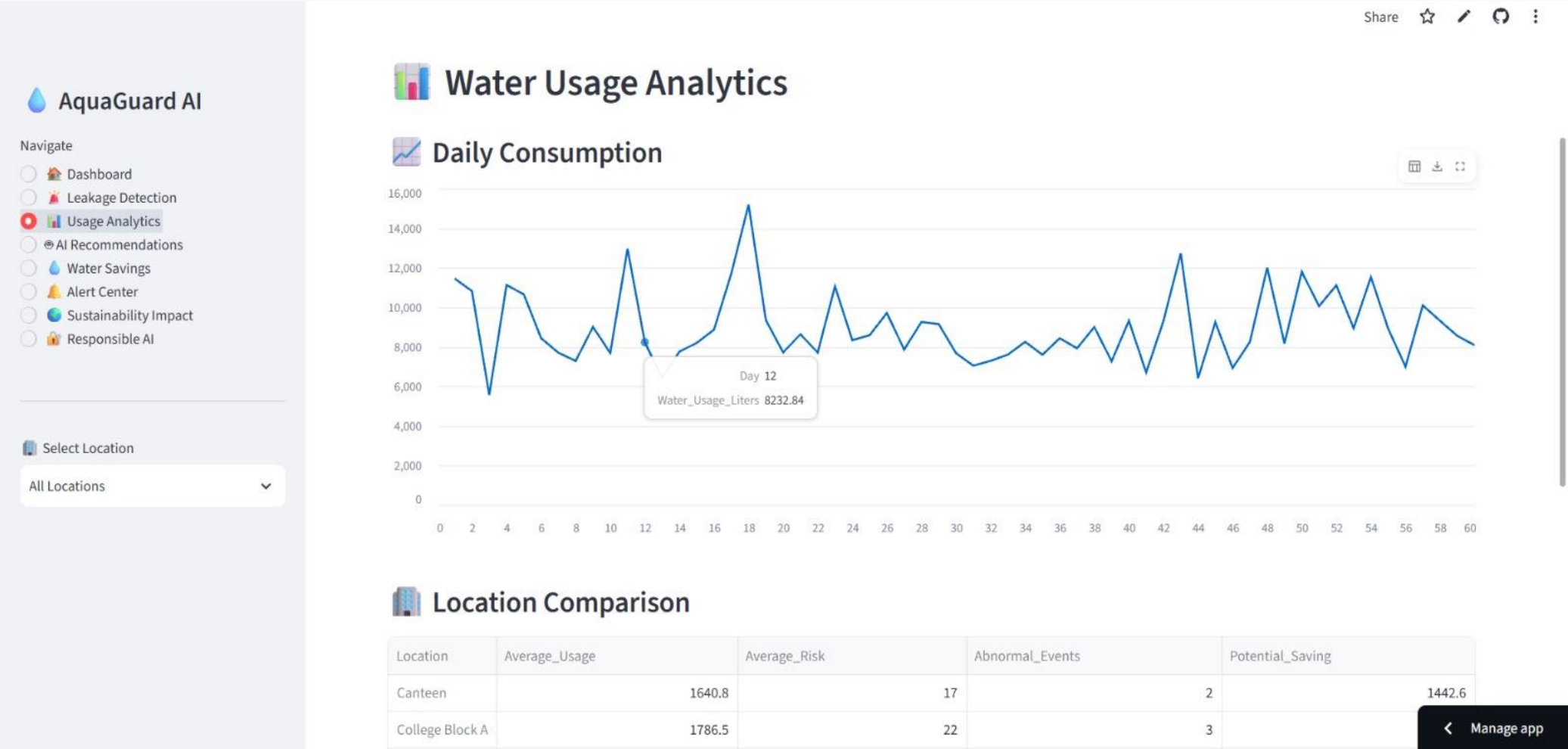
 Highest Risk Events

	Day	Location	Water_Usage_Liters	Leak_Risk_Score	Risk_Level
1	1	Hostel B	4679.57	100	 High Risk
16	4	Hostel B	5275.06	100	 High Risk
50	11	Hostel A	4962.51	100	 High Risk
238	48	College Block B	6287.83	100	 High Risk
265	54	Hostel A	6122.94	100	 High Risk
261	53	Hostel B	3573.76	100	 High Risk
246	50	Hostel B	5997.78	100	 High Risk
256	52	Hostel B	4091.61	100	 High Risk
59	12	Canteen	3077.08	100	 High Risk

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Manage app

Usage Analytics



AI Recommendations

Share

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Select Location

All Locations

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 AI Recommendations

AquaGuard AI converts detected risk patterns into practical maintenance recommendations.

 Priority Location

Hostel B | Risk Score: 100.0

 Immediate inspection recommended

- Check pipelines and water connections
- Inspect continuously running taps
- Check storage tanks and overflow
- Verify abnormal consumption at the location

 Why was this location flagged?

The system compares observed water consumption with historical normal usage patterns and calculates a risk indicator based on the deviation.

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What-if Water Saving

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Navigate

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- ☐ AI Recommendations
- ☒ Water Savings
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- ☐ Responsible AI

Select Location

All Locations

 Water Saving Analysis

Estimate how much water could potentially be saved if abnormal consumption is corrected.

 Current Potential Saving

49,593 L

 Monthly Projection

4,959 L

 Yearly Projection

59,512 L

 What-if Water Saving Simulator 

Expected reduction in abnormal consumption (%)

30

 With a 30% reduction, approximately 14,878 L could potentially be recovered from the analyzed records.

 Projected monthly saving: 1,488 L

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Alert Center

AQUAGUARD AI — PROTOTYPE OUTPUT

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Navigate

☐

 Dashboard

☐

 Leakage Detection

☐

 Usage Analytics

☐

 AI Recommendations☐☐☒☐

Select Location

All Locations 

 Sustainability Impact

 SDG 6 — Clean Water and Sanitation

AquaGuard AI supports responsible water management by identifying abnormal consumption and helping organizations take corrective action.

 Potential Saving

49,593 L

 Monthly Estimate

4,959 L

 Yearly Estimate

59,512 L

 Expected Impact

AquaGuard AI can help:

- Detect unusual water consumption earlier
- Prioritize locations requiring inspection
- Reduce avoidable water wastage
- Support data-driven maintenance decisions
- Promote responsible water consumption

 Small improvements in water-use efficiency can contribute to long-term sustainability.

 Manage app

Sustainability Impact

- Primary SDG: SDG 6 – Clean Water and Sanitation
- Potential savings are estimated from abnormal consumption.
- Expected impact: earlier detection, better prioritization and responsible water use.

Responsible AI

- Transparency: risk score is an indicator, not proof of leakage.
- Human verification is required before maintenance action.
- Water-usage data should be handled securely.
- Prototype uses simulated data and needs real-world validation.

Technology & Future

- Python • Pandas • NumPy • Scikit-learn • Streamlit • GitHub • Google Colab
- Future: IoT meter integration, real-world validation, forecasting, automated notifications and richer contextual features.

Conclusion + Demo

- Detect → Decide → Save → Sustain
- GitHub: <https://github.com/SreenivaasiniJayaraman/AquaGuard-AI->
- Live Demo: <https://eh8qnqhqmwdj4jhotfx2gm.streamlit.app/>
- Add your final Streamlit screenshots here.

